

Advanced Data Management

Propositional Logic

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Outline of the course

1. **Introduction to Propositional Logic**
 2. Introduction to First-order Logic
 3. Relational Calculus
 4. **Information Integration Systems (IIS)**
 5. Logical formalization of IISs
 6. Mapping between **Global Schema** e **Data Sources**
 7. Incomplete information databases
 8. Query answering over IISs
 9. **Ontology-Based Data Management (OBDM)**
 10. Description Logic Ontologies
 11. Query answering in ontologies
 12. Query answering in OBDM
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Formal Logic

Formal Logic

When dealing with important things in life (...like databases...) we need to agree on a way to evaluate our sentences.

- Is a database correctly representing our scenario?
- Is a tuple satisfying a given condition?

To do that, in Computer Science and Engineering we borrow constructs from a branch of mathematics and philosophy called **formal logic**.

Formal Logic:

- The **formal** study of human reasoning (**Philosophy**).
 - The use of logical formalisms to study mathematical objects (**Math**).
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Example

On a walk in the woods, you stumble upon a bridge guarded by two trolls (named **Ander** and **Bjorn**). They say:

- **Ander**: If one of us is a knight, then you can safely cross the bridge.
- **Bjorn**: Either Ander is a knight or Bjorn is a knight.

Yes... They refer to themselves in the third person... Trolls...

Questions:

1. *If both trolls are telling the truth, is it safe to cross the bridge?*
2. *If one troll lies, can we conclude that it is not safe to cross the bridge?*
3. *Can they both lie?*

Question 1

If both trolls are telling the truth, is it safe to cross the bridge?

Ander: If one of us is a knight then you can safely cross the bridge.

- If this sentence is true, to be sure we are safe we just need to find a knight!

Bjorn: Ander is a knight or Bjorn is a knight.

- If this sentence is true, we have our knight!

- **Conclusion:** you can safely cross!
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Question 2

*If **one of the trolls lies**, can we conclude that it is not safe to cross?*

- Suppose **Bjorn** lies:
 - **Bjorn** : Either Ander is a knight or Bjorn is a knight.
 - **Not Bjorn**: Ander is NOT a knight **AND** Bjorn is NOT a knight.
 - We cannot conclude that the bridge is safe to cross.
 - We cannot conclude that the bridge is not safe to cross.
 - **We do not know!**
- Suppose **Ander** lies.
 - **Ander**: If one of us is a knight, then you can safely cross the bridge.
 - **Not Ander**: One of us is a knight **AND** you can **NOT** safely cross the bridge.
 - We can conclude that it is **NOT** safe to cross!!
- If we know that Ander is lying, we can conclude that it is **not safe to cross!**

Question 3

Can both trolls lie?

Suppose **Bjorn** lies:

- **Bjorn**: Either Ander is a knight or Bjorn is a knight.
- **Not Bjorn**: Ander is **NOT** a knight **AND** Bjorn is **NOT** a knight.

Suppose **Ander** lies:

- **Ander**: If one of us is a knight, then you can safely cross the bridge.
- **Not Ander**: One of us is a knight **AND** you can **NOT** safely cross the bridge.

So, we have that **One of them is a knight** (Ander) but **none of them is** (Bjorn).

→ **Contradiction!**

Formal Logic: Some Examples

Formal logic can draw certain conclusions from certain premises.

- **Premise 1:** If Fabio is eating a croissant, then Fabio is happy.
- **Premise 2:** Fabio is eating a croissant.
- **Conclusion:** Fabio is happy!

What did we just do? **Propositional Calculus!**

- We understood (calculated) whether a fact (proposition) is true (or false) starting from the fact that some other facts are certainly true (or false)
- Under the (somewhat hidden) assumption that fact is either **true** or **false**

Truth Values

- What are the facts involved in our example?
 - Premise 1: If (Fabio is eating a croissant) then (Fabio is happy).
 - Premise 2: (Fabio is eating a croissant)
 - Conclusion: (Fabio is happy)
- At this level of abstraction, we only need to know if sentences are true
 - We call this the **truth value** of a sentence.
 - Unlike real life, we assume that every proposition is either **true** or **false**.
 - In the example, we concluded that (Fabio is Happy) is **true**.

Propositional Variables

Observe: when dealing with truth values, we do not need to know exactly what the single fact is about. We just need to know its truth value.

For this reason, to keep notation compact, instead of writing down the whole proposition, we use **Propositional Variables**.

A propositional variable is a symbol to which we associate a specific meaning.

Propositional Variables: Example

- Let's try to reason again about croissants using variables.

(Fabio is eating croissants) = C

(Fabio is happy) = H

- A straightforward translation.

Premise 1: If (Fabio is eating croissants) then (Fabio is happy).

- Premise 1: If C then H

Premise 2: (Fabio is eating a croissant)

- Premise 2: C

Conclusion: (Fabio is happy)

- Conclusion: H

Formal Logic: Analyzing Connectives

When dealing with important things in life (...like databases and croissants...) we need to draw solid conclusions

- Premise 1: **If A then D**
- Premise 2: **(A AND B) OR ((NOT B) AND D)**
- Conclusion: **D is true?**

Evaluating complex examples requires a formal definition of how propositional variables are connected to each other.

Propositional connectives: AND, OR, NOT

Propositional Logic: Syntax

Alphabets, Symbols, and Words

- An **alphabet** S is simply a countably infinite set. The elements of S are called the **symbols** of the alphabet.
- A **word** (string) w over S is defined as follows:
 - Every symbol is a string.
 - If w , w' are strings, so is $w \circ w'$
 - Nothing else is a word.
- Assume $S = \{a, b\}$
 - Words: $a, aa, ab, ba, aaa \dots$
 - The following is not a word of S : $ba**c**$

Alphabet of Propositional Formulae

- Assume a countably infinite set $R = \{p_1, p_2, \dots\}$. (Propositional Variables)
- Assume the set $O = \{\neg, \wedge, \vee\}$ (Connectives)
- Assume the set P of symbols “(” and “)”.
- Sets R , O , and P form the **alphabet** of **propositional formulae**
- Not every word in this alphabet is a formula!

Propositional Formulae

An (propositional) **atomic formula** is a (propositional) **variable**.

– For brevity, we often call atomic formulae simply **atoms**.

The set of propositional formulae $\mathcal{L}$ is the smallest such that:

1. Every atom is in $\mathcal{L}$
2. If f is in $\mathcal{L}$, then also $\neg(f)$ is in $\mathcal{L}$.
3. If f, f' are in $\mathcal{L}$, so are $(f \circ f')$, with $\circ \in \{\wedge, \vee\}$

Propositional Formulae – Example

The following are propositional formulae

1. a (atom)
2. $(a \wedge b)$
3. $\neg((a \wedge b) \vee (c \wedge d))$

The following are not propositional formulae

1. $a \wedge$
2. $\vee b$
3. $a \wedge b$ (no parenthesis, though sometimes we will omit them)

Propositional Logic: Semantics

Propositional Connectives

The set of truth values is the set $\{\mathbf{true}, \mathbf{false}\}$

To each propositional connective, we associate functions

- **Unary**: $\{\mathbf{true}, \mathbf{false}\} \rightarrow \{\mathbf{true}, \mathbf{false}\}$
- **Binary**: $\{\mathbf{true}, \mathbf{false}\} \times \{\mathbf{true}, \mathbf{false}\} \rightarrow \{\mathbf{true}, \mathbf{false}\}$

Intuitively, these functions allow us to define the **meaning** of formulae.

- Usually, we define these functions using “**truth tables**”.

AND : Truth Table

The connective **AND** requires that **both** operands must be true

AND	TRUE	FALSE
TRUE	TRUE	FALSE
FALSE	FALSE	FALSE

OR : Truth Table

The connective **OR** requires that **at least one** operand must be true

OR	TRUE	FALSE
TRUE	TRUE	TRUE
FALSE	TRUE	FALSE

NOT: Truth Table

The connective **NOT** requires that the operand **must be false**

NOT	
TRUE	FALSE
FALSE	TRUE

Propositional Interpretations

In order to understand whether a formula is *true* or *false*, we need to evaluate its variables.

A (propositional) **interpretation** is a function $I : \mathcal{L} \rightarrow \{\mathbf{true}, \mathbf{false}\}$ such that

- $I(a) \in \{\mathbf{true}, \mathbf{false}\}$, for each atom $a \in \mathcal{L}$
- $I(\neg f) = \mathbf{NOT} I(f)$
- $I(f \wedge f') = \mathbf{AND}(I(f), I(f'))$
- $I(f \vee f') = \mathbf{OR}(I(f), I(f'))$

Observe: an interpretation is uniquely determined by its interpretation of atoms

Observe: the same holds for the truth value of a formula.

Evaluation of Propositional Formulae: example

Consider an interpretation I such that:

$$I(A) = \textit{true}, \quad I(B) = \textit{false}, \quad I(C) = \textit{true}$$

and the formula $(A \wedge B) \vee ((\neg B) \wedge C)$, we have:

- $AND(I(A), I(B)) = \textit{false}$
- $NOT(I(B)) = \textit{true}$
- $AND(\textit{true}, I(C)) = \textit{true}$
- $OR(\textit{false}, \textit{true})$ is ***true***

Exercise

Considering the interpretation I defined as follows:

$$I(A) = \textit{true}, \quad I(B) = \textit{false}, \quad I(C) = \textit{true}, \quad I(D) = \textit{false}$$

What is the truth value of the following propositional formulae?

1. $(A \vee B) \wedge (\neg(C) \vee D)$

2. $\neg(A \vee B) \vee \neg(C \vee D)$

3. $(\neg(A) \wedge \neg(B)) \vee (C \vee D)$

Exercise

Considering the interpretation I defined as follows:

$$I(A) = \textit{true}, \quad I(B) = \textit{false}, \quad I(C) = \textit{true}, \quad I(D) = \textit{false}$$

What is the truth value of the following propositional formulae?

1. $(A \vee B) \wedge (\neg(C) \vee D)$ *false*

2. $\neg(A \vee B) \vee \neg(C \vee D)$ *false*

3. $(\neg(A) \wedge \neg(B)) \vee (C \vee D)$ *true*

Logical Consequence

Models and Consequences

We call a finite set of formulae a **theory**.

- Formulae in a theory are often called its **axioms**.

Assume an interpretations I , we say that

- I is a **model** of a formula f if $I(f) = \textit{true}$ (written $I \models f$).
- I is a **model** of a theory F if $I(f) = \textit{true}$, for every $f \in F$ (written $I \models F$).
- A formula f is the **consequence** of a theory G ($G \models f$) if for every interpretation I tht is a **model** of G we have that I is a **model** of f
- **Consequences** of a set of formulae A are often called the **theorems** of A .

Exercise

Which of the following are logical consequences?

1. $\{p, (\neg(p) \vee q)\} \models q$

2. $\{(p \vee q)\} \models q$

3. $\{(\neg(p) \wedge p)\} \models q$

Exercise: Solution (1)

$\{p, (\neg(p) \vee q)\} \models q$ **holds**.

Since:

1. In every model of the theory p is **true**
2. In order for $(\neg(p) \vee q)$ to be **true** then, q needs to be **true**
3. So, every model of the theory $\{p, (\neg(p) \vee q)\}$ is also a model of the formula q

Exercise: Solution (2)

$\{(p \vee q)\} \models q$ **does not hold.**

Indeed, the interpretations with $I(p) = \text{true}$, $I(q) = \text{false}$ are **models** of the theory.

Exercise: Solution (3)

$\{(\neg(p) \wedge p)\} \models q$ **holds.**

Since:

1. The set has **no model**
2. If a set has **no models**, every formula **is a consequence by definition** (**ex falso sequitur quodlibet**)

Exercise: Solution (recap)

- Which of the following are logical consequences?

1. $\{p, (\neg(p) \vee q)\} \models q$

– *True*

2. $\{p \vee q\} \models q$

– *False*

3. $\{\neg(p) \wedge p\} \models q$

– *True*

Logical Equivalences

Logical Equivalence

- Assume an interpretation I
 - I is a **model** of a formula f if $I(f) = \text{true}$ ($I \models f$).
 - I is a **model** of a theory F if $I(f) = \text{true}$, for every $f \in F$. ($I \models F$).
- A formula f is **equivalent** to a formula f' ($f \equiv f'$) if:
 - I is a model of f if and only if I is a model of f'
 - We extend this notion to theories in the natural way.
- There are several useful equivalences that one can prove.

Implication

So far, we used **if ... then** (implication) in our sentences, but we do not have a connective for that. However, we do not need one.

Indeed, **if f then g** (*f implies g*) requires that **g** is *true* whenever **f** is *true* (if **f** is *false*, then we *don't care* about **g**) also written ($f \rightarrow g$)

- **How do we translate it in our formal logic?**

When is the sentence **if f then g** true?

- If **f** is **false** then **g** can be either **true** or **false**, we don't care.
- If **f** is **true**, then **g** must be **true**.

Implication: Truth Table

<u>IMPLIES(A,B)</u>	B TRUE	B FALSE
A TRUE	TRUE	FALSE
A FALSE	TRUE	TRUE

Exercise

As we said, all the logical connectives we need are **AND**, **OR**, and **NOT**.

Claim: *IMPLIES* (f, g) is **equivalent** to *OR* $(\text{NOT}(f), g)$

$$(f \rightarrow g) \equiv (\neg(f) \vee g)$$

Prove the claim above.

Exercise 5: Solution

- As we said, all the logical connectives we need are **AND**, **OR**, and **NOT**.
- **Claim:** *IMPLIES* (f, g) is equivalent to *OR* (*NOT*(f), g)
- **Proof:** Truth table of **NOT**(f) **OR** g .

– $f = \text{true}, g = \text{true}.$	NOT (f) OR $g = \text{true}$
– $f = \text{true}, g = \text{false}$	NOT (f) OR $g = \text{false}$
– $f = \text{false}, g = \text{true}$	NOT (f) OR $g = \text{true}$
– $f = \text{false}, g = \text{false}$	NOT (f) OR $g = \text{true}$

Logical Implication: the two truth tables

<u>A IMPLIES B</u>	^B TRUE	^B FALSE	<u>NOT(A) OR B</u>	^B TRUE	^B FALSE
^A TRUE	TRUE	FALSE	^A TRUE	TRUE	FALSE
^A FALSE	TRUE	TRUE	^A FALSE	TRUE	TRUE

De Morgan's Law and Double Negation

Important equivalences for handling negation. Assume formulae f, g .

- De Morgan's Law 1: $\neg (f \wedge g) \equiv (\neg f \vee \neg g)$
 - De Morgan's Law 2: $\neg (f \vee g) \equiv (\neg f \wedge \neg g)$
 - Double Negation: $\neg \neg f \equiv f$
- **Observe:** these are not axioms. We can prove them!

Exercise

- Prove De Morgan's laws and the law of double negation.

Exercise: Solution

- Prove De Morgan's laws and the law of double negation.
- De Morgan's Law 1: $\neg (f \wedge g) \equiv (\neg f \vee \neg g)$
- **Proof:** We prove the claim showing the truth tables.

	Truth table $\neg (f \wedge g)$	Truth table $(\neg f \vee \neg g)$
• f = true , g = true .	false	false
• f = true , g = false	true	true
• f = false , g = true	true	true
• f = false , g = false	true	true

Exercise: Solution

- Prove De Morgan's laws and the law of double negation.
- Double Negation: $\neg\neg f \equiv f$
- **Proof:** We prove the claim showing the truth tables.

	Truth table f	Truth table $\neg\neg f$
• $f = \text{false}$	false	false
• $f = \text{true}$	true	true

Exercise (the trolls...again)

On a walk in the woods, you stumble upon a bridge guarded by two trolls. They say:

- **Ander**: If one of us is a knight, then you can safely cross the bridge.
 - **Bjorn**: Either Ander is a knight or Bjorn is a knight.
1. Prove, by means of propositional logic, that it is safe to cross the bridge.
 2. Prove, by means of propositional logic, that at least one of the two trolls tells the truth.

Exercise: Solution (1)

(one knight) = K , (safe bridge) = S

(Ander knight) = KA , (Bjorn knight) = KB ,

Ander: If one of us is a knight, then you can safely cross the bridge.

$\neg K \vee S$ (If K then S)

Bjorn: Either Ander is a knight or Bjorn is a knight.

$KA \vee KB$

Ander or Bjorn are knights if and only if there is a knight

$\neg(KA \vee KB) \vee K$

$\neg K \vee (KA \vee KB)$

Exercise: Solution (2)

Consider the following theory

$$\mathcal{A} = \{(\neg K \vee S), (KA \vee KB), (\neg(KA \vee KB) \vee K), (\neg K \vee (KA \vee KB))\}$$

To prove that it is safe to cross the bridge, we need to prove that $\mathcal{A} \models S$ holds

1. In each model of $\mathcal{A}$, it holds that $(KA \vee KB)$
 - because in $\mathcal{A}$ we have $(KA \vee KB)$
2. Therefore, K holds
 - because $(\neg(KA \vee KB) \vee K)$
3. Therefore, S is true
 - because $(\neg K \vee S)$
4. So, every model of $\mathcal{A}$ is also a model for S , i.e., $\mathcal{A} \models S$

Exercise: Solution (3)

In order to prove that **at least one of the two trolls tells the truth**, we show that the theory in which both are lying has **no model**

Let consider the negations of the troll's statements

$$\neg(\neg K \vee S) \equiv (K \wedge \neg S)$$

$$\neg(KA \vee KB) \equiv (\neg KA \wedge \neg KB)$$

Consider the theory $\mathcal{A}'$ obtained from the above statements:

$$\mathcal{A}' = \{(K \wedge \neg S), (\neg KA \wedge \neg KB), \neg(KA \vee KB) \vee K, \neg K \vee (KA \vee KB)\}$$

Claim: The theory $\mathcal{A}'$ has no model

- By contradiction, suppose such model exists. Call it I
- Since $I((K \wedge \neg S)) = \text{true}$ we have $I(K) = \text{true}$
- Since $I((\neg KA \wedge \neg KB)) = \text{true}$ then $I((KA \vee KB)) = \text{false}$,
- Since $I((KA \vee KB)) = \text{false}$ and $I(\neg K \vee (KA \vee KB)) = \text{true}$ then $I(\neg K) = \text{true}$ contradicting the fact that I is a (propositional) interpretation.

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.
